school boards vote on subjects to be taught in the classroom. In some cases, votes are cast according to the whims of cultural, political, or religious tides. Around the world, varying belief systems lead to political differences that are not always resolved peacefully. The power and beauty of physical laws is that they apply everywhere, whether or not you choose to believe in them.

In other words, after the laws of physics, everything else is opinion.

Not that scientists don't argue. We do. A lot. But when we do, we typically express opinions about the interpretation of insufficient or ratty data on the bleeding frontier of our knowledge. Wherever and whenever a physical law can be invoked in the discussion, the debate is guaranteed to be brief: No, your idea for a perpetual motion machine will never work; it violates well-tested laws of thermodynamics. No, you can't build a time machine that will enable you to go back and kill your mother before you were born—it violates causality laws. And without violating momentum laws, you cannot spontaneously levitate and hover above the ground, whether or not you are seated in the lotus position.†

Knowledge of physical laws can, in some cases, give you the confidence to confront surly people. A few years ago I was having a hot-cocoa nightcap at a dessert shop in Pasadena, California. Ordered it with whipped cream, of course. When it arrived at the table, I saw no trace of the stuff. After I told the waiter that my cocoa had no whipped cream, he asserted I couldn't see it because it sank to the bottom. But whipped cream has low density, and floats on all liquids that humans consume. So I offered the waiter two possible explanations: either somebody forgot to add the whipped cream to my hot cocoa or the universal laws of physics were different in his restaurant. Unconvinced, he defiantly brought over a dollop of whipped cream to demonstrate his claim. After bobbing once or twice the whipped cream rose to the top, safely afloat.

What better proof do you need of the universality of physical law?

† You could, in principle, perform this stunt if you managed to let forth a powerful and sustained exhaust of flatulence.